

```
#install.packages("readxl")
library(readxl)
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(readr)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats   1.0.0      v stringr   1.5.1
v ggplot2   3.5.1      v tibble    3.2.1
v lubridate 1.9.3      v tidyr     1.3.1
v purrr     1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Description of dataset:

What is the data source? (1-2 sentences on where the data is coming from, dates included, etc.) How does the dataset relate to the group problem statement and question?

The data comes from CalEnviroScreen 4.0 (2021) by California's Office of Environmental Health Hazard Assessment (OEHHA) and the California Department of Health Care Access and Information (HCAI), which provides county-level asthma emergency department visit rates from 2016–2021. These datasets would help us explore how environmental burdens across California counties relate to asthma outcomes and identify which areas could benefit from targeted interventions that could make the biggest impact.

Identify data types for 5+ data elements/columns/variables:

5 columns - chhs_asthma_ed: county, ED visit rate (and # ED visits?) - ces_measures: county, asthma, PM2.5, diesel.PM, poverty, pollution burden.

Questions we have: - in ces_measures, what does the asthma percentile column represent? numbers seem high to be a rate of the general population.

Describe cleaning that each data source may need

cleaning: - NA values from ces_measures: asthma, number_of_ed_visits, poverty: We will replace with 0, "Unknown", or median(df\$column1, na.rm = TRUE) depending on context.

We changed the census_tract column of the ces_measures to character to account for any leading zeros.

```
ces_datadictionary <- read_excel("~/PHW251_2025/251-Group11/calenviroscreen_datadictionary.x
```

New names:

```
* `` -> `...2`
```

```
head(ces_datadictionary)
```

```
# A tibble: 6 x 2
  `Table 1`      ...2
  <chr>         <chr>
1 Data Dictionary for CES4.0FINAL_results Data Spreadsheet <NA>
2 Variable Name      Description
3 Census Tract       Census Tract ID from~
4 Total Population   2019 ACS population ~
```

5 California County
6 ZIP

California county th~
Postal ZIP Code that~

You can add options to executable code like this

```
ces_measures <- read.csv("~/PHW251_2025/251-Group11/calenviroscreen_measures_2021.csv")

ces_measures <- clean_names(ces_measures)

ces_measures <- ces_measures %>% mutate(census_tract = as.character(census_tract))

#this is to show the type of each element
str(ces_measures%>% select(california_county, asthma, pollution_burden_score, pm2_5, diesel_
```

```
'data.frame':  8035 obs. of  6 variables:
 $ california_county      : chr  "Fresno" "San Joaquin" "Los Angeles" "Fresno" ...
 $ asthma                 : num  129.5 105.9 76.1 139.4 139.1 ...
 $ pollution_burden_score: num  9.64 8.97 9.48 8.28 8.16 ...
 $ pm2_5                  : num  13.9 11.9 12.3 13.5 13.8 ...
 $ diesel_pm              : num  1.123 0.538 0.781 0.174 1.39 ...
 $ poverty                : num  76 73.2 62.6 65.7 72.7 43.5 79.5 56.8 72.2 78.8 ...
```

```
head(ces_measures)
```

	census_tract	california_county	ozone	ozone_pctl	pm2_5	pm2_5_pctl
1	6019001100	Fresno	0.06031100	82.47666	13.90635	97.71002
2	6077000700	San Joaquin	0.04588368	44.97822	11.88408	72.59490
3	6037204920	Los Angeles	0.04791979	53.72744	12.25164	89.20971
4	6019000700	Fresno	0.06031100	82.47666	13.52094	95.88052
5	6019000200	Fresno	0.06031100	82.47666	13.81896	97.47355
6	6037542402	Los Angeles	0.04259906	29.89421	12.25258	89.22215
	diesel_pm	diesel_pm_pctl	drinking_water	drinking_water_pctl	lead	
1	1.1227118	98.71811	733.9469	84.38866	89.60085	
2	0.5381054	91.15121	389.8466	41.55114	77.30227	
3	0.7808326	96.55258	787.9403	92.53153	92.56366	
4	0.1738153	57.08774	733.9469	84.38866	68.38508	
5	1.3896581	99.30305	733.9469	84.38866	75.41454	
6	0.5197984	90.64095	459.7366	55.31410	91.08803	
	lead_pctl	pesticides	pesticides_pctl	tox_release	tox_release_pctl	traffic
1	96.47133	1.0019247	42.89655	4859.095	92.19805	1037.0957
2	86.83050	63.1325743	73.65517	519.628	52.35059	856.3959

3	98.39950	0.0000000	0.00000	3682.693	87.73443	2522.6223
4	77.00063	44.5748736	71.58621	1630.343	74.93123	690.5022
5	85.11657	16.6254961	64.41379	1975.208	78.95724	909.6509
6	97.49212	0.4067027	36.06897	12491.996	98.36209	1480.5210
	traffic_pctl	cleanup_sites	cleanup_sites_pctl	groundwater_threats		
1	60.3750	70.50	98.18962	54.25		
2	48.2875	61.90	97.46174	78.60		
3	92.8375	38.75	92.96379	20.50		
4	35.2750	16.50	77.32363	9.50		
5	52.2625	10.50	62.44867	28.25		
6	77.6625	19.40	81.29899	53.30		
	groundwater_threats_pctl	haz_waste	haz_waste_pctl	imp_water_bodies		
1		91.16719	3.10	96.30708	0	
2		95.08551	1.27	88.57710	13	
3		68.91914	11.62	99.65586	7	
4		44.82816	2.36	94.12310	0	
5		78.05081	0.35	56.39974	0	
6		91.00116	2.72	95.34083	10	
	imp_water_bodies_pctl	solid_waste	solid_waste_pctl	pollution_burden		
1		0.00000	6.00	79.95418	78.97740	
2		91.87316	9.25	89.27835	73.43499	
3		66.73667	4.85	73.08133	77.67318	
4		0.00000	5.75	78.14433	67.84684	
5		0.00000	0.00	0.00000	66.82595	
6		82.96934	5.50	77.61741	76.43661	
	pollution_burden_score	pollution_burden_pctl	asthma	asthma_pctl		
1		9.643125	99.92533	129.54	97.22084	
2		8.966398	99.29060	105.88	94.21735	
3		9.483880	99.86310	76.10	82.76421	
4		8.284085	97.39888	139.45	98.24277	
5		8.159435	96.93839	139.08	98.16800	
6		9.332895	99.77598	86.77	88.59671	
	low_birth_weight	low_birth_weight_pctl	cardiovascular_disease			
1		7.80	95.62484	21.47		
2		6.88	88.72209	20.26		
3		7.11	90.89043	20.87		
4		10.65	99.78188	22.68		
5		10.25	99.66641	22.64		
6		6.28	80.03593	22.20		
	cardiovascular_disease_pctl	education	education_pctl	linguistic_isolation		
1		92.24826	44.5	93.22956	16.0	
2		88.12313	46.4	94.46975	29.7	
3		90.15454	52.2	97.39307	17.1	

4	94.56630	41.4	90.93900	15.7
5	94.42921	43.6	92.57150	20.0
6	93.66899	33.1	83.57378	NA
	linguistic_isolation_pctl	poverty	poverty_pctl	unemployment
1	79.37475	76.0	98.91960	12.8
2	95.53390	73.2	98.39196	19.8
3	81.55366	62.6	93.39196	6.4
4	78.71160	65.7	95.35176	15.7
5	86.56110	72.7	98.30402	13.7
6	NA	43.5	74.04523	9.3
	unemployment_pctl	housing_burden	housing_burden_pctl	pop_char
1	93.83134	30.3	91.03929	93.15511
2	99.20614	31.2	92.28137	93.16541
3	61.53045	20.3	63.96705	83.75181
4	97.34513	35.4	96.41318	94.64123
5	95.28891	32.7	94.15716	95.39887
6	83.22488	23.7	76.34981	83.36615
			pop_char_score	pop_char_pctl
			9.663213	99.72264
			9.664281	99.73525
			8.687785	95.78921
			9.817371	99.88654
			9.895964	99.94957
			8.647779	95.46142

```
#showing the summary statistics
summary(ces_measures$asthma)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
4.28	30.06	45.71	51.98	65.80	243.29	11

```
unique(ces_measures$california_county)
```

[1] "Fresno"	"San Joaquin"	"Los Angeles"	"Stanislaus"
[5] "San Bernardino"	"Kern"	"Riverside"	"Madera"
[9] "Kings"	"Sacramento"	"Merced"	"San Diego"
[13] "Solano"	"Alameda"	"Orange"	"Tulare"
[17] "Contra Costa"	"Imperial"	"Yolo"	"Sutter"
[21] "Santa Clara"	"San Francisco"	"Ventura"	"Yuba"
[25] "San Mateo"	"Monterey"	"Butte"	"Santa Cruz"
[29] "Santa Barbara"	"Glenn"	"Sonoma"	"Humboldt"
[33] "Lake"	"Mendocino"	"Marin"	"Tehama"
[37] "Nevada"	"Napa"	"San Benito"	"Placer"
[41] "Colusa"	"Shasta"	"Modoc"	"Lassen"
[45] "Siskiyou"	"Del Norte"	"Tuolumne"	"Mariposa"
[49] "Amador"	"San Luis Obispo"	"Calaveras"	"El Dorado"
[53] "Inyo"	"Plumas"	"Trinity"	"Sierra"
[57] "Mono"	"Alpine"		

```
summary(ces_measures$poverty)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
1.00	16.30	27.70	31.34	44.30	96.70	75

```
summary(ces_measures$diesel_pm)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000052	0.067683	0.144929	0.225442	0.286794	14.611221

```
summary(ces_measures$pm2_5)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.875	8.575	10.119	10.153	11.938	16.395

```
summary(ces_measures$pollution_burden_score)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5595	4.0679	5.1893	5.2125	6.3205	10.0000

The `echo: false` option disables the printing of code (only output is displayed).

```
ces_scores_demog <- read.csv("~/PHW251_2025/251-Group11/calenviroscreen_scores_demog_2021.csv")
```

```
ces_scores_demog <- clean_names(ces_scores_demog)
```

```
head(ces_scores_demog)
```

	census_tract	ces_4_0_score	ces_4_0_percentile	ces_4_0_percentile_range
1	6001400100	4.847988	2.798790	1-5% (lowest scores)
2	6001400200	4.882107	2.874433	1-5% (lowest scores)
3	6001400300	11.204244	15.935451	15-20%
4	6001400400	12.390190	18.973777	15-20%
5	6001400500	16.734731	29.740292	25-30%
6	6001400600	20.024315	37.594554	35-40%

	total_population	children_10_years	pop_10_64_years	elderly_64_years	hispanic
1	3120	7.82	66.12	26.06	3.78
2	2007	10.46	66.32	23.22	8.67

3	5051	11.42	73.04	15.54	6.95
4	4007	9.38	78.79	11.83	12.10
5	4124	9.12	81.96	8.92	9.46
6	1745	9.97	81.43	8.60	7.51

	white	african_american	native_american	asian_american	other_multiple
1	74.26	3.43	0.00	12.53	5.99
2	73.49	2.59	0.20	8.52	6.53
3	67.99	9.09	0.00	12.14	3.84
4	63.74	6.64	0.87	10.48	6.16
5	45.44	21.39	0.00	11.34	12.37
6	49.28	20.52	0.17	10.54	11.98

	county
1	Alameda County
2	Alameda County
3	Alameda County
4	Alameda County
5	Alameda County
6	Alameda County

```
chhs_asthma_ed <- read.csv("~/PHW251_2025/251-Group11/chhs_asthma_ed.csv")
chhs_asthma_ed <- clean_names(chhs_asthma_ed)
str(chhs_asthma_ed)%>% select(county, number_of_ed_visits, age_adjusted_ed_visit_rate))
```

```
'data.frame':  4484 obs. of  3 variables:
 $ county      : chr  "CALIFORNIA" "ALAMEDA" "ALPINE" "AMADOR" ...
 $ number_of_ed_visits : int  191904 9939 0 196 1044 185 97 6858 140 592 ...
 $ age_adjusted_ed_visit_rate: num  50.4 64.3 0 58.4 50.2 48 41.4 65.2 53 36.4 ...
```

```
head(chhs_asthma_ed)
```

	county	year	strata	strata_name	age_group	number_of_ed_visits
1	CALIFORNIA	2015	Total population	All ages	All ages	191904
2	ALAMEDA	2015	Total population	All ages	All ages	9939
3	ALPINE	2015	Total population	All ages	All ages	0
4	AMADOR	2015	Total population	All ages	All ages	196
5	BUTTE	2015	Total population	All ages	All ages	1044
6	CALAVERAS	2015	Total population	All ages	All ages	185
	age_adjusted_ed_visit_rate					
1	50.4					
2	64.3					

3	0.0
4	58.4
5	50.2
6	48.0

```
summary(chhs_asthma_ed$number_of_ed_visits)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.0	43.5	191.0	2013.5	852.0	191904.0	681

5 columns - chhs_asthma_ed: county, ED visit rate (and # ED visits?) -

ces_measures: county, asthma, PM2.5, diesel.PM, poverty, pollution burden.

In order to prepare each dataset for analysis (milestone #3), you will be expected to:

- **Cal Enviro Screen datasets**
 - **Recode common variable(s) to be combined with other datasets**
 - * this means make column names match?
 - **Identify environmental measure(s) of interest and summarize into a county-level value**
 - * for indicators that are broken apart by census tract, combine into one county-level number, one row per county
 - we could do median of pm2.5, diesel.PM, poverty, pollution burden
 - **Summarize CalEnviroScreen score into a county level variable**
 - * ces_scores_demog dataframe: ces_4_0_score column- combine into one row per county averaging all of the values.
 - one row per county, mean of all ces_4_0_score values.
 - **NOTE: the methods for each summary variable (environmental measure and CalEnviroScreen score) should be different (i.e. if you calculate the mean for one, please use another method for the other). If you look at more than one environmental measure you can re-use summarization methods for this.**
- **Asthma ED dataset - Angela**
 - Recode common variable(s) to be combined with other datasets__ *done*

- Recode values that are reading in incorrectly__ *done*
- Subset to most recent year and county level data__ *done*
- Select demographic strata of interest__ *demographic strata of interest included was total population, so that we can more readily compare the numbers across data sets.*
- Pivot table to only include one row per county __ *used other functions to create one row per county.*

For all scenarios, please turn in an html document created from an Rmd or Qmd with the following components:

- Subset rows or columns, as needed
- Create new variables needed for analysis (minimum 2)
 - New variables should be created based on existing columns; for example
 - * Calculating a rate
 - * Combining character strings
 - * Aggregation
- Clean variables needed for analysis (minimum 2)
 - Examples
 - * Recode invalid values
 - * Handle missing data
 - * Recode categories
- Data re-structured as needed (aggregated/summarized and/or pivoted)
- Data dictionary based on clean dataset (minimum 4 data elements), including:
 - Variable name
 - Data type
 - Description
- One or more tables with descriptive statistics for 4 data elements
- Html output that is professionally prepared for presentation
 - Only the necessary information is outputted (you should suppress, for example, entire data frame outputs)
 - Use of headers and sub headers to create an organized document

Milestone 3 Rubric

		Milestone 3 Rubric
Criteria	Ratings	Pts
This criterion is linked to a Learning Outcome Subset rows or columns as needed	2 ptsFull Marks	2 pts
This criterion is linked to a Learning Outcome Create new variables needed for analysis (minimum 2)	3 ptsFull Marks	3 pts
This criterion is linked to a Learning Outcome Clean variables needed for analysis (minimum 2)	3 ptsFull Marks	3 pts
This criterion is linked to a Learning Outcome Restructure datasets, as needed	2 ptsFull Marks	2 pts
This criterion is linked to a Learning Outcome Data dictionary based on clean dataset (minimum 4 data elements)	4 ptsFull Marks	4 pts
This criterion is linked to a Learning Outcome One or more tables with descriptive statistics for 4 data elements	4 ptsFull Marks	4 pts
This criterion is linked to a Learning Outcome HTML document that is professionally prepared for presentation	2 ptsFull Marks	2 pts

Total Points: 20

Milestone #3 starts here

Data Dictionary

California County	California county that the census tract falls within
Mean CES 4.0 Score	Mean of CalEnviroScreen Score, Pollution Score multiplied by Population Characteristics Score

Median PM2.5	Annual median PM 2.5 concentrations
--------------	-------------------------------------

Median Diesel PM	Annual median Diesel PM emissions from on-road and non-road sources
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Median Poverty	Median Percent of population living below two times the federal poverty level by county
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Median Traffic	Median traffic density, in vehicle-kilometers per hour per road length, within 150 meters of the census tract boundary by county
----------------	--

Cleaning /Creating New Variables in Dataframes

```
#angela asthma dataframe
df_asthma <- chhs_asthma_ed %>%
  clean_names() %>%
  filter(county != "CALIFORNIA") %>%
  filter(strata == "Total population") %>%
  filter(year == 2020) %>%
  select(county, year, number_of_ed_visits, age_adjusted_ed_visit_rate) %>%
  mutate(county = str_to_title(county)) %>%
  mutate(county = paste0(county, " County")) %>%
  mutate(across(where(is.numeric), ~if_else(is.na(.), 0, .)))
df_asthma
```

	county	year	number_of_ed_visits	age_adjusted_ed_visit_rate
1	Alameda County	2020	4282	25.8
2	Alpine County	2020	0	0.0
3	Amador County	2020	103	31.2
4	Butte County	2020	474	24.2
5	Calaveras County	2020	111	30.4
6	Colusa County	2020	49	24.0
7	Contra Costa County	2020	3040	27.5
8	Del Norte County	2020	104	40.6

9	El Dorado County	2020	351	20.6
10	Fresno County	2020	3504	34.0
11	Glenn County	2020	55	19.5
12	Humboldt County	2020	443	36.0
13	Imperial County	2020	474	25.1
14	Inyo County	2020	47	28.3
15	Kern County	2020	2532	27.7
16	Kings County	2020	426	27.1
17	Lake County	2020	269	46.8
18	Lassen County	2020	90	33.4
19	Los Angeles County	2020	25920	26.2
20	Madera County	2020	475	30.8
21	Marin County	2020	280	11.7
22	Mariposa County	2020	31	22.8
23	Mendocino County	2020	315	39.8
24	Merced County	2020	1034	37.1
25	Modoc County	2020	43	57.3
26	Mono County	2020	40	33.3
27	Monterey County	2020	917	21.3
28	Napa County	2020	322	22.7
29	Nevada County	2020	174	21.6
30	Orange County	2020	4870	15.9
31	Placer County	2020	623	16.7
32	Plumas County	2020	57	42.2
33	Riverside County	2020	5888	24.9
34	Sacramento County	2020	5489	35.7
35	San Benito County	2020	219	36.5
36	San Bernardino County	2020	6763	31.1
37	San Diego County	2020	5749	17.5
38	San Francisco County	2020	1437	16.6
39	San Joaquin County	2020	2945	38.6
40	San Luis Obispo County	2020	432	18.3
41	San Mateo County	2020	1082	14.0
42	Santa Barbara County	2020	680	15.5
43	Santa Clara County	2020	2814	14.5
44	Santa Cruz County	2020	458	17.3
45	Shasta County	2020	402	24.8
46	Sierra County	2020	0	0.0
47	Siskiyou County	2020	86	24.8
48	Solano County	2020	1770	41.7
49	Sonoma County	2020	884	18.8
50	Stanislaus County	2020	1804	32.8
51	Sutter County	2020	183	18.8

52	Tehama County 2020	173	28.0
53	Trinity County 2020	27	28.5
54	Tulare County 2020	1135	23.8
55	Tuolumne County 2020	129	28.6
56	Ventura County 2020	1394	17.4
57	Yolo County 2020	456	22.8
58	Yuba County 2020	191	25.0

```
#dineli environmental measures dataset clean up
df_env_measures <- read_csv("calenviroscreen_measures_2021.csv", show_col_types = FALSE) %>%
  clean_names() %>%

  # recode common variable(s) for combining with other datasets
  mutate(county = str_to_title(california_county)) %>%
  mutate(county = paste0(county, " County")) %>%

  # recode values that are reading incorrectly
  mutate(across(where(is.numeric), ~na_if(., -999))) %>%
  mutate(across(where(is.numeric), ~na_if(., -1))) %>%

  # identify and select environmental measures of interest
  select(county, pm2_5, diesel_pm, poverty, traffic) %>%

  # group by county and summarize to one row per county (median)
  group_by(county) %>%
  summarise(
    median_pm2_5 = median(pm2_5, na.rm = TRUE),
    median_diesel_pm = median(diesel_pm, na.rm = TRUE),
    median_poverty = median(poverty, na.rm = TRUE),
    median_traffic = median(traffic, na.rm = TRUE),
    .groups = "drop"
  ) %>%

  # replace missing numeric values with 0 (to match your asthma dataset syntax)
  mutate(across(where(is.numeric), ~if_else(is.na(.), 0, .)))

df_env_measures
```

A tibble: 58 x 5

	county	median_pm2_5	median_diesel_pm	median_poverty	median_traffic
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Alameda County	8.71	0.260	17.2	928.

2	Alpine County	3.05	0.00262	38.9	70.2
3	Amador County	8.24	0.0122	23.1	213.
4	Butte County	8.45	0.0703	36	455.
5	Calaveras County	8.44	0.00744	26.9	215.
6	Colusa County	7.78	0.0349	37.5	135.
7	Contra Costa Cou~	8.78	0.151	16.3	892.
8	Del Norte County	5.75	0.0231	43.8	265.
9	El Dorado County	7.63	0.0204	20.4	435.
10	Fresno County	13.5	0.109	47.6	567.

i 48 more rows

```
#angela CES scores data frame
```

```
df_scores <- ces_scores_demog %>%
  group_by(county) %>%
  summarise(mean_ces_4_0_score = mean(ces_4_0_score, na.rm = TRUE))

df_scores
```

```
# A tibble: 58 x 2
  county          mean_ces_4_0_score
  <chr>          <dbl>
1 Alameda County    22.9
2 Alpine County     13.6
3 Amador County     20.7
4 Butte County      21.7
5 Calaveras County  16.1
6 Colusa County     27.0
7 Contra Costa County 21.0
8 Del Norte County  21.4
9 El Dorado County   10.2
10 Fresno County    40.9
# i 48 more rows
```

Combine Dataframes and Re-Structured

```
df_combined <- df_scores %>%
  left_join(df_asthma, by = "county") %>%
  left_join(df_env_measures, by = "county")

df_combined
```

```
# A tibble: 58 x 9
  county      mean_ces_4_0_score year number_of_ed_visits age_adjusted_ed_visi~1
  <chr>          <dbl> <dbl>          <dbl>          <dbl>
1 Alameda ~      22.9  2020          4282          25.8
2 Alpine C~      13.6  2020           0           0
3 Amador C~      20.7  2020          103          31.2
4 Butte Co~      21.7  2020          474          24.2
5 Calavera~      16.1  2020          111          30.4
6 Colusa C~      27.0  2020           49           24
7 Contra C~      21.0  2020         3040          27.5
8 Del Nort~      21.4  2020          104          40.6
9 El Dorad~      10.2  2020          351          20.6
10 Fresno C~      40.9  2020         3504          34
# i 48 more rows
# i abbreviated name: 1: age_adjusted_ed_visit_rate
# i 4 more variables: median_pm2_5 <dbl>, median_diesel_pm <dbl>,
#   median_poverty <dbl>, median_traffic <dbl>
```

Create a Table for HTML Output

```
library(gt)
#Create a clean table for html output
df_combined %>%

  gt() %>%
  tab_header(
    title = "Merged County-Level Dataset",
    subtitle = "CalEnviroScreen Scores, Environmental Indicators, and Asthma ED Visits (2020)"
  ) %>%
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) %>%
  tab_options(
```

```

    table.width = pct(100),
    table.font.size = px(13)
  )

```

Data Dictionary

```

#Data Dictionary
library(gt)

data_dictionary <- tibble::tibble(
  variable = c(
    "county",
    "mean_ces_4_0_score",
    "median_pm2_5",
    "median_diesel_pm",
    "median_poverty",
    "median_traffic"
  ),
  type = c("character", "numeric", "numeric", "numeric", "numeric", "numeric"),
  description = c(
    "California county.",
    "Mean CalEnviroScreen 4.0 score (Pollution Burden × Population Characteristics, tract → county median).",
    "Annual median PM2.5 concentration (tract → county median).",
    "Annual median Diesel PM index from on-/off-road sources (tract → county median).",
    "Median % of population below 2× federal poverty level (tract → county median).",
    "Median traffic density (veh-km/hour per road length within 150m of tract), tract → county median."
  )
)

data_dictionary %>%
  gt() %>%
  tab_header(title = "Data Dictionary - Clean County-Level Dataset")

```


Merged County-Level Dataset

CalEnviroScreen Scores, Environmental Indicators, and Asthma ED Visits (2020)

county	mean_ces_4_0_score	year	number_of_ed_visits	age_adjusted_ed_visit_rate	medicaid_rate
Alameda County	22.91	2,020.00	4,282.00	25.80	25.80
Alpine County	13.62	2,020.00	0.00	0.00	0.00
Amador County	20.74	2,020.00	103.00	31.20	31.20
Butte County	21.70	2,020.00	474.00	24.20	24.20
Calaveras County	16.11	2,020.00	111.00	30.40	30.40
Colusa County	26.99	2,020.00	49.00	24.00	24.00
Contra Costa County	21.03	2,020.00	3,040.00	27.50	27.50
Del Norte County	21.37	2,020.00	104.00	40.60	40.60
El Dorado County	10.17	2,020.00	351.00	20.60	20.60
Fresno County	40.92	2,020.00	3,504.00	34.00	34.00
Glenn County	27.11	2,020.00	55.00	19.50	19.50
Humboldt County	18.55	2,020.00	443.00	36.00	36.00
Imperial County	40.42	2,020.00	474.00	25.10	25.10
Inyo County	14.76	2,020.00	47.00	28.30	28.30
Kern County	37.03	2,020.00	2,532.00	27.70	27.70
Kings County	41.42	2,020.00	426.00	27.10	27.10
Lake County	21.38	2,020.00	269.00	46.80	46.80
Lassen County	17.29	2,020.00	90.00	33.40	33.40
Los Angeles County	37.98	2,020.00	25,920.00	26.20	26.20
Madera County	37.82	2,020.00	475.00	30.80	30.80
Marin County	9.93	2,020.00	280.00	11.70	11.70
Mariposa County	17.24	2,020.00	31.00	22.80	22.80
Mendocino County	19.80	2,020.00	315.00	39.80	39.80
Merced County	44.81	2,020.00	1,034.00	37.10	37.10
Modoc County	18.86	2,020.00	43.00	57.30	57.30
Mono County	13.35	2,020.00	40.00	33.30	33.30
Monterey County	21.00	2,020.00	917.00	21.30	21.30
Napa County	18.43	2,020.00	322.00	22.70	22.70
Nevada County	12.61	2,020.00	174.00	21.60	21.60
Orange County	23.61	2,020.00	4,870.00	15.90	15.90
Placer County	11.75	2,020.00	623.00	16.70	16.70
Plumas County	16.00	2,020.00	57.00	42.20	42.20
Riverside County	26.78	2,020.00	5,888.00	24.90	24.90
Sacramento County	25.24	2,020.00	5,489.00	35.70	35.70
San Benito County	25.77	2,020.00	219.00	36.50	36.50
San Bernardino County	33.65	2,020.00	6,763.00	31.10	31.10
San Diego County	19.98	2,020.00	5,749.00	17.50	17.50
San Francisco County	18.31	2,020.00	1,437.00	16.60	16.60
San Joaquin County	38.29	2,020.00	2,945.00	38.60	38.60
San Luis Obispo County	13.62	2,020.00	432.00	18.30	18.30
San Mateo County	16.84	2,020.00	1,082.00	14.00	14.00
Santa Barbara County	19.52	2,020.00	680.00	15.50	15.50
Santa Clara County	17.04	2,020.00	2,814.00	14.50	14.50
Santa Cruz County	15.53	2,020.00	458.00	17.30	17.30
Shasta County	17.16	2,020.00	402.00	24.80	24.80
Sierra County	18.28	2,020.00	0.00	0.00	0.00
Siskiyou County	18.55	2,020.00	86.00	24.80	24.80
Solano County	24.78	2,020.00	1,770.00	41.70	41.70
Sonoma County	14.88	2,020.00	884.00	18.80	18.80
Stanislaus County	38.84	2,020.00	1,804.00	32.80	32.80
Sutter County	31.70	2,020.00	183.00	18.80	18.80
Tehama County	26.03	2,020.00	173.00	28.00	28.00
Trinity County	14.15	2,020.00	27.00	28.50	28.50
Tulare County	42.46	2,020.00	1,135.00	23.80	23.80
Tuolumne County	18.77	2,020.00	129.00	28.60	28.60
Ventura County	20.80	2,020.00	1,394.00	17.40	17.40
Yolo County	22.53	2,020.00	456.00	22.80	22.80

Data Dictionary — Clean County-Level Dataset

variable	type	description
county	character	California county.
mean_ces_4_0_score	numeric	Mean CalEnviroScreen 4.0 score (Pollution Burden $\times$ Population C
median_pm2_5	numeric	Annual median PM2.5 concentration (tract $\rightarrow$ county median).
median_diesel_pm	numeric	Annual median Diesel PM index from on-/off-road sources (tract $\rightarrow$
median_poverty	numeric	Median % of population below $2\times$ federal poverty level (tract $\rightarrow$ co
median_traffic	numeric	Median traffic density (veh-km/hour per road length within 150m o